VF Asia Footwear Laboratory Testing Manual

Test Method for Leather Sampling	**Test method # VF-H02** **REF: TBLQA-LX-006** **Page 1 of 5**
Date Issued : August 25, 2014 **Revision # : 1** **Reason for revision: VF Aligned with TBL Test method**	**Created by: Nila Babia Erazo / Chris Adam.**

PURPOSE AND SCOPE: To define the requirements for incoming leather inspection, sample selection, cutting and testing of all leather used in the manufacture of VF footwear.

REFERENCES: ASTM D2813

MATERIALS: All leather

PROCEDURE

1. Incoming Inspection:

Factory incoming leather inspection (production and samples) should include the following criteria:

1.1 Verification of completed VF Footwear Brands – Vans, Reef, TNF, and TBL Certificate of Analysis.
1.2 Color & Character (surface look/feel/break): compare to associated approved TBL Leather Standard swatch
1.3 Weight/Thickness
1.4 Finish Adhesion (Tape)
1.5 Temper (softness/firmness)
1.6 Selection/Grading
1.7 Area Measurement

2. Testing Requirements:

2.1 For every leather shipment larger than 40 square feet, all suppliers must supply a completed Certificate of Analysis for every lot.
2.2 Factories must perform testing on all leather receipts as per the sample selection plan outlined in Section 3.
2.3 Waterproof Testing (Waterproof leather only)
2.3.1.1 Maeser Flex.
2.3.1.2 Dynamic Absorption
2.3.1.3 Static Absorption

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2.4. Physical Testing

2.4.1 Wet/Dry Crocking

2.4.2 Mullen burst or Lastometer

2.4.3 Tensile Strength and Elongation

2.4.3 UV Resistance (black leather only)

2.5. Test results for each drum (color) lot within the shipment must be held for a minimum of six months from date of shipment and made available to Timberland upon request.

3. Sample selection:

3.1 Samples should be selected at Random based on the square footage of the Shipment Drum/Lot as outline in Table I.

3.2 Factories must take cuttings from different bundles/boxes/pallets to best represent the shipment (i.e. when directed to take 3 cuttings, select one from the first, middle, and last bundle/ box/ pallet).

3.3 If the Random Sample Selection fails, move to the Expanded Sample Selection criteria, using different sides of leather. Failing the Expanded testing would constitute that lot as sub-standard.

3.4 Sub-standard material must be put in a discrepant location and the supplier notified immediately. The following information should be reported to the supplier: leather type, purchase order number, date received, lot identification number (if available), square footage of shipment, and test data.

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Table I. Sample Selection Plan for testing of incoming leather.

Shipment/Drum Lot Square Footage	Random (Factory)		Expanded (Factory)		Initial (Tannery)	
	# of Cuttings	Pass Rate	# of Cuttings	Pass Rate	# of Cuttings	Pass Rate
0 to 5000	1	1 of 1	3	3 of 3	3 per drum lot	3 of 3
5,001 to 10,000	2	2 of 2	6	5 of 6	6 per drum lot	5 of 6
10,001 to 15,000	3	3 of 3	9	8 of 9	9 per drum lot	8 of 9
15,001 to 20,000	3	3 of 3	9	8 of 9	9 per drum lot	8 of 9
20,001 to 25,000	5	4 of 5	15	12 of 15	15 per drum lot	12 of 15
25,001 to 30,000	5	4 of 5	15	12 of 15	15 per drum lot	12 of 15
30,001 to 35,000	5	4 of 5	15	12 of 15	15 per drum lot	12 of 15
35,001 to 40,000	5	4 of 5	15	12 of 15	15 per drum lot	12 of 15
40,001 to 45,000	5	4 of 5	15	12 of 15	15 per drum lot	12 of 15
45,001 & up	10	8 of 10	20	16 of 20	20 per drum lot	16 of 20

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4. Cutting Selection

4.1 All cuttings must be taken at least 18 hours prior to testing. This is to allow the "airing-off" of any remaining solvents, which may cause failure.

4.2 If the moisture content of the test specimen(s) is greater than 15%, mechanical drying may be used to bring the moisture content within an approximate range of 12-14%. Caution is necessary so as not to over dry the cuttings. Over dried cuttings can fail due to damage to the fiber structure while flexing. Cuttings should not be exposed to temperatures in excess of 200°F/93°C for a duration longer than 30 minutes.

4.3 Cuttings for waterproof and physical testing should be taken from the locations outlined in Table II and Fig. 1.

Table II. Cutting locations for waterproof and physical testing.

	Test Specimen		
Sample Type	**Location 1**	**Location 2**	**Location 3**
Random (Factory)	6" (15cm) down from backbone 8" (20cm) in from butt		
Expanded (Factory)	6" (15cm) down from backbone 8" (20cm) in from butt		
Initial (Tannery)	6" (15cm) down from backbone 8" (20cm) in from butt	10" (25cm) down from backbone, mid-side cut	10" (25cm) down from backbone, mid shoulder cut

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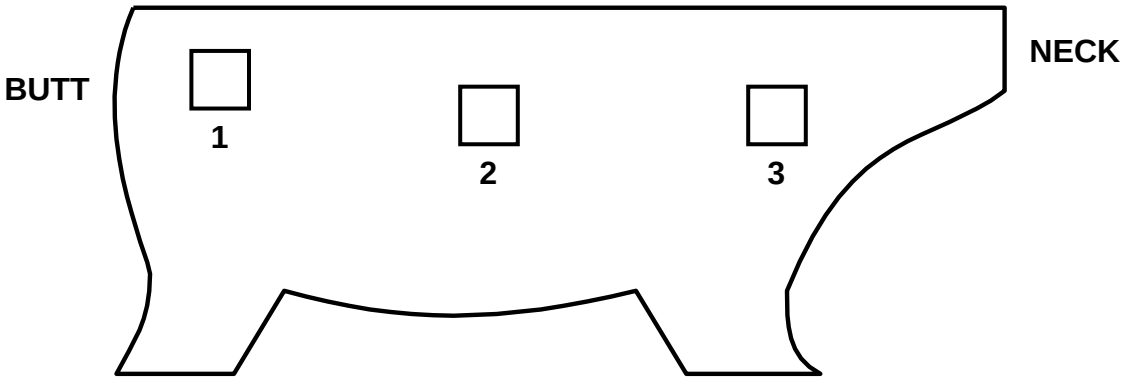


Figure 1. Cutting locations for waterproof and physical testing.

SUPPORT DOCUMENTATION

VF-E02	Colorfastness to Crocking
VF-A03	Gauge
VF-E11	Maeser Water Penetration and Dynamic Water Absorption Test
VF-BO2	Mullen Burst
VF-E10	Static Water Absorption
VF-F05	Surface Finish Leather – Scotch Tape Test
VF-B01	Tensile – Leather
VF-A06	Leather Softness /emper – Leather
VF-E03	UV Aging – Color Changes in Leather

TBLQA-LX-001	Leather Grading Parameters
TBLQA-LX-008	Leather Area Measurement Verification